Space Technology Students' Society

Quantum Computing in Practice: A Review of Real-World Applications

Ishan Krish, Raghav Toshniwal, Shradha

Abstract: The goal of this paper's thorough literature review on the various applications of quantum computing is to analyze and summarize the various new developments in multiple interdisciplinary fields. This review aims to delve into the theoretical aspects of quantum computing, such as superposition, entanglement, qubit manipulation, and other fundamental elements, to develop a deep understanding of its applications. This paper examines various research papers and publications to create a review that uncovers their uses in real-world applications. 6G internet, algorithmic optimization, cryptocurrencies, semiconductor engineering, and transportation systems are a few to name. The review also identifies the challenges faced in these fields and what the future holds, highlighting the technological gaps and the upcoming developments to fill the void. Ultimately, this review provides a deep understanding of the current standpoint of quantum computing and sheds light on the most promising future directions.

Keywords: Quantum computing,Quantum cryptography, Quantum Key Distribution,Post-Quantum Cryptography, Quantum algorithms, Quantum healthcare, Quantum machine learning, Quantum error mitigation, Quantum Internet, Quantum error correction, Quantum chemistry, Quantum semiconductors, Quantum Arts and Humanities, Quantum transportation modelling, Quantum Space Technology

1 INTRODUCTION

Quantum computing is one of the driving forces behind a computational revolution, which is characterised by exponential technological advancement. Quantum computing refers to computational prowess that utilises concepts of superposition, entanglement, and interference in quantum mechanics to manipulate data in ways much different from a classical computer. This shift into quantum computing is extremely time-saving and resource-friendly. But first, let us look at a few essential terms to understand the basics.

1. **Qubit:** Qubit, or a quantum bit, is the basic unit of quantum information. Unlike a classical bit, it can simultaneously be a superposition of the two states (0 and 1).

2. **Superposition:** A quantum system can exist in multiple states simultaneously, allowing quantum computers to perform various tasks in parallel.

3. **Entanglement:** A unique phenomenon dealing with the linking of qubits such that the state of a particular qubit is dependent on the other and vice versa. It is very crucial for quantum communication and algorithms.

4. **Quantum Gate:** A quantum gate performs a similar task to logic gates in classical circuits, i.e., they are responsible for the change in state of the qubits.

5. **Quantum Circuit:** It is just a collection or sequence of multiple quantum gates.

Now, apart from the theoretical aspect. Quantum computing has been gaining popularity in almost every field. Its uses are numerous, and expanding quickly from 6G internet and optimization algorithms to biosciences and space technology. They have even found their applications in machine learning and game theory. However, these are just a few applications to name. Now that we grasp the basics, let's dive into the details. This literature review examines the state of quantum computing applications today, examining how this game-changing technology has and can influence multiple fields, challenging how classical methods approach these problems.

2 QUANTUM INTERNET

The union of the 6G communication systems with quantum computing (QC) is a revolutionary technological shift that points towards increased security, efficiency, and computational power. Over the coming years, a connection forged between the two technologies is poised to bring the human, digital, and physical realms much closer together.

2.1 The Key Fundamental

6G requires the formation of Key Performance Indicators (KPIs) and Key Value Indicators (KVIs). A KPI is a performance measure over a given time, while a KVI indicates the value created by each function for future use. In the context of Quantum Internet, the KVI generally refers to the QKD or the Quantum Key Distribution, which employs quantum mechanics to share cryptographic keys for network establishment.

2.2 Computational Benefits of Quantum Systems

A Quantum Computer is exponentially faster than a Classical Computer in executing various tasks (such as factoring a large number), reducing energy expenditure, and aiding in achieving higher efficiency and better-performing 6G networks.

2.3 Optimising Network Design

- The design of a network communication is vital. Quantum algorithms can be used to optimize a communication network design and operation that can lead to environment and resource-friendly systems. These designs have also been found to support a high level of simulation capabilities that, in turn, help test the 6G architecture.

- Quantum networking is also key to dispersing and distributing quantum computing, which allows multiple quantum computers to work on the same algorithm, furthering their computation prowess.

- This can be achieved by linking them through quantum channels, supported by quantum repeaters. Quantum repeaters, often placed in between the connections, are crucial for enabling long-distance quantum communication by setting up entanglement between distant nodes.

2.4 Conclusion

The concept of Quantum Internet is still quite new, but it promises to bring a major change in the world of computing and communication, along with the provision of enhanced security.

3 QUANTUM CRYPTOGRAPHY

3.1 Introduction

Cryptography makes sure that only parties that have exchanged keys can read the encrypted message (also called authentic parties). Quantum computers threaten the main goal of every secure and authentic communication because they are able to do computations that classical (conventional) computers cannot. Consequently, quantum computers can break the cryptographic keys quickly by calculating or searching exhaustively all secret keys, allowing an eavesdropper to intercept the communication channel between authentic parties (sender/receiver). This task is considered to be computationally infeasible by a conventional computer. [6] [7]

3.2 Current Research Scenario

To date, There has been lot of research on this topic :

1. **Quantum Cryptography: Public Key Distribution and Coin Tossing (Bennett & Brassard, 1984):** Introduces the BB84 protocol, the first quantum key distribution (QKD) method, using quantum mechanics to securely share cryptographic keys. It shows how quantum states detect eavesdropping, forming the basis of quantum cryptography.

2. **The Security of Quantum Key Distribution (Mayers, 2001):**Provides a formal proof of BB84s unconditional security against all attacks, including those from quantum computers. It examines practical QKD imperfections, ensuring robust security analysis.

3. **Post-Quantum Cryptography (Bernstein, Buchmann, Dahmen, 2009):** Explores quantum-resistant cryptographic schemes (PQC), including lattice-based, code-based, hash-based, and multivariate polynomial methods, designed to counter quantum attacks like Shors algorithm.

4. **Quantum Cryptography Beyond Quantum Key Distribution (Broadbent & Schaffner, 2016):**

Surveys advanced quantum cryptographic primitives like quantum money, digital signatures, and multi-party computation, highlighting quantum computings role in enabling innovative protocols.

5. **Shors Algorithm and the Threat to Classical Cryptography (Shor, 1997)** Introduces Shors algorithm, which efficiently factors numbers and solves discrete logarithms on quantum computers, breaking classical systems like RSA and ECC, necessitating quantum-safe cryptography.

6. **Device-Independent Quantum Cryptography (Pironio et al., 2010):** Presents device-independent QKD, securing communication without trusting devices, using Bell inequalities to counter quantum computing-based attacks.

7. **Quantum-Safe Cryptography: NISTs Post-Quantum Cryptography Standardization (Alagic et al., 2022):** Summarizes NISTs selection of quantum-resistant algorithms (e.g., CRYSTALS-Kyber), addressing security and implementation challenges against quantum attacks.

8. **Quantum Cryptography with Continuous Variables (Grosshans et al., 2003)** Explores continuous-variable QKD using coherent states, offering scalable alternatives to discrete-variable QKD, compatible with quantum computing systems.

9. **Quantum Cryptography in the Noisy-Intermediate-Scale Quantum (NISQ) Era (Fernández-Caramés & Fraga-Lamas, 2020):** Examines QKD and PQC feasibility with NISQ quantum computers, addressing practical challenges in implementing quantum cryptography.

3.2.1 Quantum Key Distribution (QKD)

Quantum Key Distribution (QKD) is a method to securely exchange cryptographic keys between two parties, Alice and Bob, using quantum mechanics principles like Heisenbergs uncertainty and entanglement. It complements, rather than replaces, classical cryptography by ensuring secure key sharing, not encryption/decryption. Popular protocols include:

- BB84 (1984, Bennett & Brassard): Uses two bases (Rectilinear, Diagonal) with four quantum states to encode bits. Alice sends qubits, Bob measures in random bases, and they sift the key by comparing bases, achieving a 50% sifted key with a 25% initial error rate. Security is tested via QBER, followed by error correction and privacy amplification to eliminate eavesdropper (Eve) information.

- B92 (1992, Bennett): Simplified BB84, using two non-orthogonal states. Bob measures in one basis, retaining bits where hes certain of the state, followed by QBER-based security checks.

- E91 (1991, Ekert): Entanglement-based, using entangled photon pairs. Alice and Bob measure in random bases, splitting results into compatible (key) and incompatible (eavesdropper detection) groups, ensuring security via Bells inequality.

- Six State Protocol (SSP): Extends BB84 with three bases (six states), increasing Eves error rate to 66%, improving eavesdropper detection but requiring quantum memory.

- SARG04: Like B92, uses two non-orthogonal states but encodes bases in additional qubits, avoiding basis revelation, and resists PNS attacks.

- COW: Encodes bits in time slots with light pulses, using decoy states for security, suitable for high bit rates.

Protocol	Year	Author(s)	Type	Principle Used	Number of states	Sifting phase	Detection of Eavesdropper	Vulnerabilities
BB84	1984	C. H. Bennett and G. Brassard	Prepare and Measure	Heisenberg's Uncertainty Principle	4	Revealing bases	QBER	PNS, IRA, MITM, DoS attacks
B92	1992	C. H. Bennett	Prepare and Measure	Heisenberg's Uncertainty Principle	2	Alice = 1 - Bob	QBER	PNS, IRA, DoS attacks
E91	1991	Artur Ekert	Entanglement Based	Quantum Entanglement	2	Using Bell's Inequality	Bell's Inequality	DoS attack
SSP	1999	H. Bechmann-Pasquinucci and N. Gisin	Prepare and Measure	Heisenberg's Uncertainty Principle	6	Revealing bases	QBER	PNS, MITM, DoS attacks
SARG04	2004	V. Scarani, A. Acín, G. Ribordy and N. Gisin	Prepare and Measure	Heisenberg's Uncertainty Principle	4	Revealing non-orthogonal states	QBER	PNS, DoS attacks
COW	2004	D. Stucki, S. Fasel, N. Gisin, Y. Thoma and H. Zbinden	Entanglement Based	Heisenberg's Uncertainty Principle	Time slots	Revealing the times	Break of coherence of successive pulses	DoS attack

Fig. 1 Table 1: A comparison of QKD Protocols [**Mavroeidis2018**]

3.2.2 Security Threats:

QKD faces attacks like Photon Number Splitting (PNS), Intercept and Resend (IRA), Man-in-the-Middle (MITM), Denial-of-Service (DoS), side-channel, and Trojan Horse attacks. Mitigation includes decoy states, avoiding public basis disclosure, and dedicated networks. Protocols

like BB84, B92, E91, and SARG04 are proven unconditionally secure, assuming Eve is limited only by quantum mechanics, while Alice and Bob use current technology.

3.3 Quantum Computing Algorithms

3.3.1 Shors Algorithm in Asymmetric Cryptography

(Asymmetric Cryptography uses a pair of keys, a public key to encrypt and a private key to decrypt. It enables secure communication without needing to share a secret key in advance.) Shors algorithm can make modern asymmetric cryptography collapse since it is based on large prime integer factorization or the discrete logarithm problem. Shors algorithm exploits quantum parallelism and Fourier transforms to factor numbers and solve discrete logarithms, posing a significant threat to classical asymmetric cryptography and driving the need for quantum-safe alternatives.

3.3.2 Grovers algorithm in Symmetric Cryptography

(Symmetric Cryptography uses the same secret key for both encrypting and decrypting data. Its fast and efficient, but requires that both parties securely share the key beforehand.) Grovers algorithm is used to search unsorted databases. The algorithm can find a specific entry in an unsorted database of square root of N entries in N searches. In comparison, a conventional computer would need N/2 searches to find the same entry. Bone and Castro remarked on the impact of a possible application of Grovers algorithm to crack the Data Encryption Standard (DES), which relies its security on a 56-bit key. The authors remarked that the algorithm needs only 185 searches to find the key.

3.3.3 Asymmetric Encryption Schemes Affected

All public key algorithms used today are based on two mathematical problems, the aforementioned factorisation of large numbers (e.g., RSA) and the calculation of discrete logarithms (e.g., DSA signatures and ElGamal encryption). Both have similar mathematical structures and can be broken with Shors algorithm rapidly.

Recent algorithms based on **elliptic curves** (such as ECDSA) use a modification of the discrete logarithm problem that makes them equally weak against quantum computers. Kirsch and Chow mentioned that a modified Shors algorithm can be used to decrypt data encrypted with ECC. In addition, they emphasized that the relatively small key space of ECC compared to RSA makes it easier to be broken by quantum computers.

Furthermore, Proos and Zalka explained that 160-bit elliptic curves could be broken by a 1000-qubit quantum computer, while factorising 1024-bit RSA would require a 2000-qubit quantum computer. The number of qubits needed to break a cryptosystem is relative to the algorithm proposed. In addition, they show in some detail how to use Shors algorithm to break ECC over GF(p). On the other hand, Grovers algorithm is a threat only to some symmetric cryptographic schemes. NIST points out that if the key sizes are sufficient, symmetric cryptographic schemes (specifically the Advanced Encryption Standard- AES) are resistant to quantum computers. Another aspect to be taken into consideration is the robustness of algorithms against quantum computing attacks, also known as quantum cryptanalysis.

3.3.4 Hash Functions

The family of hash functions suffers from a similar problem as symmetric ciphers since their security depends on a fixed output length. Grovers algorithm can be utilized to find a collision in a hash function in square root steps of its original length (it is like searching an unsorted database). To provide a b-bit security level against Grovers quantum algorithm, a hash function must provide at least a 3b-bit output. As a result, many of the present hash algorithms are disqualified for use in the quantum era. However, both SHA-2 and SHA-3 with longer outputs remain quantum resistant.

3.3.5 Post quantum cryptography

PQC research builds on the threat posed by quantum algorithms (e.g., Shors) to classical cryptography, necessitating quantum-resistant alternatives. QKD extends quantum mechanics to secure communication, while PQC schemes aim for practical deployment on classical systems. NISTs standardization process and NSAs migration plans to quantum-safe or post-quantum cryptographic algorithms underscore the urgency of transitioning to quantum-safe cryptography, positioning PQC as a critical response to quantum advancements.

Post-Quantum Cryptography (PQC) focuses on developing cryptographic systems secure against quantum and classical computers, interoperable with existing protocols. Key works, such as NISTs 2016 call for quantum-resilient algorithms, highlight 82 proposals (59 encryption/key ex-

change, 23 signatures), with ongoing evaluation to establish standards.

QKD vs PQC Quantum Key Distribution (QKD), introduced by Bennett and Brassards BB84 protocol in 1984, leverages quantum mechanics for secure key exchange using prepare-and-measure (e.g., BB84) or entanglement-based (e.g., E91) approaches, exploiting properties like Heisenbergs uncertainty or entanglement, with families like discrete variable, continuous variable, and distributed phase reference coding. In contrast, Post-Quantum Cryptography (PQC) schemes, including lattice-based (e.g., NTRU), multivariate-based, hash-based (e.g., Lamport, Merkle), and code-based (e.g., McEliece) systems, rely on mathematical problems resistant to quantum attacks like Shors algorithm, unlike vulnerable RSA or ECC.

Despite progress, QKD faces hardware vulnerabilities, such as Lydersens 2010 blinding attack, necessitating countermeasures like Yuans photocurrent monitoring or Los decoy states for photon number splitting attacks, while PQC schemes struggle with large key sizes (NTRU, McEliece), encryption security issues (multivariate), or one-time use limitations (hash-based), highlighting the need for compact, efficient, and secure algorithms to address these trade-offs and hardware flaws.

PQC systems

- Lattice-based (e.g., NTRU, GGH): Rely on hard lattice problems like the Short Vector Problem, effective in high dimensions.
- Code-based (e.g., McEliece): Use error-correcting codes with intentional errors for encryption, decoded by legitimate users.
- Multivariate (e.g., Rainbow): Based on solving multivariate polynomial equations, used for encryption and signatures.
- Hash-based (e.g., XMSS, SPHINCS): Generate one-time signatures via hash trees, secure but limited to a single use. These approaches avoid quantum-vulnerable problems, but face efficiency trade-offs (e.g., large key sizes in McEliece) and implementation challenges.

Conclusion about PQC In todays world, where information plays a particularly important role, the transmission and storage of data must be maximally secure. Quantum computers pose a significant risk to both conventional public key algorithms (such as RSA, ElGamal, ECC, and DSA) and symmetric key algorithms (3DES, AES). Year by year, it seems that we are getting closer to creating a fully operational universal quantum computer that can utilize strong quantum algorithms such as Shors algorithm and Grovers algorithm.

The consequence of this technological advancement is the absolute collapse of the present public key algorithms that are considered secure, such as RSA and Elliptic Curve Cryptosystems. The answer to that threat is the introduction of cryptographic schemes resistant to quantum computing, such as quantum key distribution methods like the BB84 protocol, and mathematical-based solutions like lattice-based cryptography, hash-based signatures, and code-based cryptography.

3.4 Current Work

Numerous implementations of QKD exist today. The University of Cambridge, in collaboration with Toshiba, implemented the BB84 protocol over a 100 km fibre link in 2008. The longest and the highest bit rate for distances over 100 km was achieved in 2015 by the University of Geneva and Corning Inc. In 2020, DRDO in India also tested its home-grown QKD setup. As the existence of optical links connecting every communicating party is highly impractical, there have been efforts to perform QKD over free space.

However, this type of communication requires suitable atmospheric conditions, and the Signal to Noise Ratio (SNR) might, at times, be too low for practical usage. Nevertheless, free-space QKD experiments have been successfully conducted in Los Alamos and China. Very recently, the Indian Space Research Organisation used quantum-key encrypted signals for video conferencing. Companies such as QNu Labs based out of Bengaluru, MagiQ Technologies Inc. from New York, and ID Quantique from Geneva have started offering commercial QKD systems.

Other companies that include HP, IBM, Toshiba, and Mitsubishi, are actively performing research to create QKD systems. Although quantum mechanics is mostly used for key distribution, research around the world is being focused on quantum digital signatures, public key encryption, quantum fingerprinting, and quantum random number generators.

3.5 Open Challenges

It has already been stated that QKD over optical fibre links quickly becomes inefficient. To overcome this problem, research is being focused on free-space QKD. However, it comes with its problems, such as the requirement for almost ideal conditions. Although there have been great efforts to improve QKD, it is only being used by some governmental organisations and research institutes. With that being said, the QKD networks that exist today do not offer very high speeds. Low-cost QKD devices that are available for the general public are a distant dream.

Most quantum cryptographic protocols make the assumption that all communicating parties have ideal single photon sources, which is hardly the case in practicality. Although numerous QKD protocols have been proposed and their security has been proven, it is hard to compare them with one another. QKD devices available today behave differently than theoretically predicted due to unavoidable imperfections. Such imperfections enable side-channel attacks. Device-independent QKD has been proposed to avoid such attacks. This is a relatively new field of research that is yet to take off completely.

In spite of the security QKD offers, it is still required to verify the identities of the communicating parties before key distribution. There are no concrete solutions to this problem. Apart from QKD, other applications of quantum mechanics in cryptography are yet to become mainstream. As stated earlier, quantum cryptography offers unconditional security. Although this property could make quantum cryptosystems highly preferable, it could also mean genuine parties such as crime investigation departments and intelligence agencies will find it hard to break them.

4 QUANTUM COMPUTING IN SEMICONDUCTOR

A quantum semiconductor is a material that shows both semiconductor properties and quantum mechanical effects. This is mostly owing to their small size. For example, gate-defined quantum dots are a family of semiconductor-based quantum dots where the confinement of electrons (or holes) is obtained through electrostatic gating in combination with physical or band-structure confinement, and where qubit states can be encoded in the charge or spin degrees of freedom, or combinations thereof. [2]

4.1 Superconducting Qubits

Semiconductor qubits are extremely versatile in quantum applications and computation. It has been observed that gate-controlled qubits have longer coherence times and a fault-tolerant possibility, which makes them viable for quantum computation. Coherence time refers to the time period during which a wave maintains the same phase.

These semiconductor qubits can be artificially engineered by configuring quantized electronic states by exploiting the nuclear or electronic degrees of freedom.

- In various quantum sensing applications, the observable qubit is made dependent on its environment.
- The qubits take on the task as nodes for quantum communication, enabling non-classical communication.

Gate-controlled quantum dots are optimised, and their spins are manipulated with slow and fast gate voltages. Voltage-gated channels refer to the two types of inactivation that prevent the channels from reopening after a period.

There also exist quite a few types of qubits involved in semiconductors and nanotechnology.

- Singlet-triplet Qubit: The qubit is encoded in the singlet and unpolarized triplet states of two electrons in a double quantum dot.[5] It basically involves encoding quantum information in the combined spin state of two electrons that are spatially separated from each other.
- Exchange-only Qubit: This qubit uses three electrons in a triple quantum dot, providing two axes of rotation via gate-controlled voltage
- Charge-Spin Hybrid Qubit: As the name suggests, this qubit is a hybrid that combines the advantages of its charge-like nature (speed) and spin-like nature (coherence)
- Multi-dot Spin Qubit: It is a solid-state qubit where quantum information is encoded in the spin states of multiple electrons present in the quantum dots.

4.2 Gate Teleportation Between Distant Superconducting Chips

Building on the knowledge of semiconductor qubits, it has been discovered that there is a possibility of deterministic teleportation of quantum states and entangling gates between distant superconducting chips.

Firstly, two superconducting quantum chips are linked via a cryogenic microwave line, enabling real-time quantum state and gate teleportation. The architecture integrates tunable couplers for high-fidelity entanglement distribution. [19] The controlled-NOT gate achieves entanglement fidelity, which basically means a secure connection between the two chips. Through coherence, the entanglement established carries the information through the channel. The entanglement established carries the information through the channel. This protocol requires only one entangled pair and 2 bits, minimizing resource overhead for modular quantum computing [19] Decoherence effects are suppressed through the application of optimized pulse sequences, which completes the connection, enabling stable and high-fidelity gate teleportation over a long-distance superconducting link

4.3 Conclusion

The intersection of microelectronics and quantum computing promises to bring a revolutionary change in multiple domains, given their wide usage and computational dominance.

5 QUANTUM HEALTHCARE

Quantum computing has several possible benefits over classical computing in biomedical problems that have high computational complexity (Table 1). For example, QCs can be used to optimize solutions for complex problems that have an inherently underlying quantum structure, such as simulating molecular interactions and protein folding Similarly, quantum algorithms, such as Grovers algorithm, can search large, unstructured datasets more efficiently than their classical counterparts, enabling more effective optimization of complex problems in healthcare and biomedical research.

However, there are also notable drawbacks, including current limitations in error correction, hardware scalability, and the need for specialized programming skills to effectively use quantum computing resources.

Artificial intelligence and machine learning Quantum computing brings novel algorithms for machine learning applications, with recent publications demonstrating potential advantages for feature selection and classification models.

Genomics Quantum computing may help with the analysis of large-scale genomic data, leading to improved understanding of genetic diseases and personalized medicine. Quantum algorithms like Grovers search algorithm and its variations may be employed to search and analyze vast genomic databases more efficiently than classical methods

Protein folding Understanding protein folding is essential for the study of diseases related to protein misfolding, such as Alzheimers and Parkinsons. Quantum computing can potentially improve the efficiency of predicting protein folding patterns. Algorithms like the Quantum Approximate Optimization Algorithm (QAOA) can be used to tackle optimization problems, including protein folding.

Drug discovery Quantum computing could significantly accelerate the drug discovery process by simulating molecular interactions and predicting the behavior of drug candidates. Quantum algorithms like the Variational Quantum Eigensolver (VQE) and Quantum Phase Estimation Algorithms (QPEA) can be used to determine molecular ground-state energies, which is crucial for understanding molecular properties and reactions.

Network analysis Network analysis in systems biology, utilized extensively in bioinformatics, involves tasks like molecular modeling and mapping complex biological pathways. Quantum computing has been applied to enhance community detection within networks, offering approaches that are comparable to classical methods. This has been described using quantum annealing and other quantum-based algorithms, which show promise in effectively partitioning networks into communities, a critical step in understanding biological and computational systems more deeply.

Cryptography Quantum computing poses a significant challenge to current encryption methods like RSA, as quantum algorithms, such as Shors algorithm, can efficiently factor large prime numbers, potentially undermining these encryption schemes 58. Conversely, it also enables the development of quantum-resistant encryption techniques, such as Quantum Key Distribution (QKD), which may enhance modern cybersecurity.

In **machine learning and artificial intelligence**, quantum variants of classical modelssuch as quantum neural networks and support vector machineshave been developed and tested on healthcare datasets, including medical imaging and electronic health records. While these models currently perform comparably to classical counterparts, they present potential for future speed and scal-

ability improvements, particularly in feature selection and data dimensionality reduction through quantum annealing.

In **genomics**, quantum algorithms have been used for tasks like DNA sequence alignment, variant detection, and phylogenetic tree construction. These efforts show that QC can effectively manage the complexity of genomic datasets, albeit presently limited to small-scale implementations due to hardware constraints.

The field of **protein folding** has also benefited from quantum approaches, where quantum optimization algorithms have been applied to predict molecular conformations. These early studies suggest QC may enable more efficient energy minimization and structure prediction than classical methods, supporting its promise in understanding diseases associated with protein misfolding.

Drug discovery workflows have incorporated quantum algorithms to predict binding affinities, molecular docking configurations, and to generate novel molecular structures. Though primarily conducted in simulation or simplified systems, these applications point to the long-term potential of QC to accelerate and refine drug development pipelines. Quantum methods have additionally been tested in biological network analysis, particularly in the context of community detection within complex biological systems. Here, quantum annealing has demonstrated performance on par with, or in some cases superior to, classical techniques.

Lastly, in the domain of **cryptography and data security**, QC presents a dual challenge and opportunity. On one hand, quantum algorithms like Shors pose a threat to current encryption standards; on the other, quantum-resistant cryptographic methods are being explored to secure sensitive healthcare data against future quantum-enabled breaches.

Taken together, the studies underscore QCs emerging role as a complementary tool in **biomedical computing**. While real-world impact remains limited by technical maturity, ongoing interdisciplinary efforts are laying the groundwork for more scalable and practical quantum applications in healthcare.

5.1 Using computers to ESKAPE the antibiotic resistance crisis

The next leap toward improving antibiotic lead discovery rates is tied directly to computational efficacy because this step currently bottlenecks CADD(computer aided drug design) processes; Fig shows how a quantum computing system is expected to be applied with respect to a drug discovery pipeline, and how it would be implemented directly to CADD techniques. Algorithms listed under the Fault Tolerant Quantum Computing (FTQC) side only will require error-free calculation to be successfully implemented. [9]

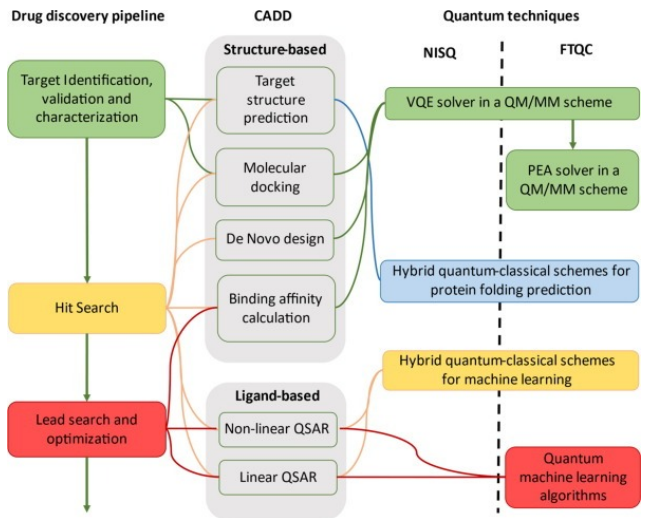

Fig. 2 Quantum computing systems within the drug discovery pipeline application considering multiple structure- and ligand-based approaches. Abbreviations: CADD, computer-aided drug design; FTQC, Fault Tolerant Quantum Computing; QSAR, quantitative structureactivity relationship; MM, XXX; NISQ, Noisy Intermediate Scale Quantum; QM, quantum mechanics; VQE, Variational Quantum Eigensolver. Adapted from [9]

5.2 Quantum Computing for Quantum Chemistry and Drug Discovery

The primary goal in quantum ground-state calculations is to determine the systems lowest possible energy. This is typically done by initializing a parameterized guess of the wave function and optimizing those parameters to minimize the system's energy. However, the computational complexity of such tasks increases rapidly with system size, challenging classical computing approaches.

Quantum computers offer a promising alternative by leveraging qubit superposition and coherence. The ability to maintain qubits in a superposed state allows for highly parallelized and complex computations. Yet, limitations such as short coherence times, limited qubit numbers, and

low fidelity in current hardware restrict large-scale quantum many-body simulations.

To address these constraints, **hybrid quantum-classical algorithms** have been developed. These split tasks between quantum and classical processors, allowing quantum circuits to operate within coherence time limits while classical processors iteratively update system parameters. Among such algorithms, the **Variational Quantum Eigensolver (VQE)** stands out as a key method. VQE applies a unitary operator to a trial wave function and evaluates the energy expectation value on a quantum processor. A classical optimizer (e.g., gradient descent) then adjusts parameters to minimize the energy, enabling efficient modeling of ground and excited states in quantum systems. VQEs practicality for quantum chemistry has been demonstrated in early studies, including bond energy calculations in hydrogen molecules.

Beyond VQE, **Quantum Phase Estimation (QPE)** is another essential algorithm used to compute eigenvalues of unitary matrices, helping solve the Schrödinger equation for quantum systems. QPE can be implemented via classical post-processing (e.g., using Kitaevs algorithm) or through inverse quantum Fourier transforms using techniques such as phase kickback. However, like VQE, QPE is currently limited by hardware constraints, including qubit coherence and circuit depth.

To improve the success rate of QPE, recent work has focused on better initial guesses for wave functions. For example, Sugisaki et al. developed a method using **configuration state functions (CSFs)** instead of traditional Slater determinants to initialize wave functions. They demonstrated this approach on systems such as hydrogen chains, nitrogen trimers, and methane bond cleavage.

Comparatively, VQE is more feasible on todays Noisy Intermediate Scale Quantum (NISQ) devices because it operates with shorter coherent evolution. QPE, while theoretically more precise, requires deeper circuits and longer coherence, making it less practical on current hardware. Both methods, however, are essential for ab initio quantum chemistry, and their computational demands and implementation strategies are active areas of research.

Despite hardware constraints, progress continues. For instance, Smith et al. used a 20-qubit IBM device to study quantum particle transport, revealing meaningful physical behavior even though absolute accuracy was low, partly due to the daily recalibration needs of the device. To further explore large, complex chemical systemsespecially in **drug discovery**future development hinges on moving beyond the NISQ era to **fault-tolerant quantum computing (FTQC)**. FTQC would enable the execution of deeper, more accurate quantum algorithms like VQE and QPE without current fidelity and coherence constraints. With such capabilities, quantum computing could significantly accelerate computational drug discovery by improving tasks like energy minimization, target interaction modelling, and prediction of molecular binding.

Notably, researchers at Google AI Quantum recently applied quantum methods to study hydrogen chains and the isomerization of diazene. Using VQE along with **error mitigation** techniques, they achieved **chemical accuracy,** suggesting potential scalability for larger systems. This milestone confirms the feasibility of applying quantum computing to real-world quantum chemistry problems and indicates that with continued algorithmic and hardware development, quantum computing could revolutionize in **silico drug discovery.**

Ultimately, the integration of high-fidelity hardware, advanced quantum algorithms, and robust error mitigation will be critical. If successful, quantum computing could overcome longstanding limitations in modelling molecular interactions, bonding, and target flexibilityareas where classical methods often fall short. This would provide powerful new tools for studying complex biological interactions (e.g., proteinprotein, drugDNA), and enable rapid, precise modelling of treatments for issues like antibiotic resistance, transforming the future of biomedical research and pharmaceutical innovation.

5.3 A new quantum-enhanced approach to AI-driven medical imaging systems

Medical Imaging Systems (MIS) play a crucial role in modern medicine by providing accurate diagnostic and treatment capabilities. These systems use various physical processes to create images inside the human body for healthcare professionals to identify and address medical conditions. There is a growing interest in integrating artificial intelligence (AI) in medicine from various sources recently. Presently, with improved algorithms and more significant availability of training data, AI can help or even replace some of the tasks that were being performed by medical professionals.

Typically, most MIS performance enhancements are

achieved by leveraging transistor-based technologies. However, such implementations showcase certain disadvantages: for instance, slow processing speeds, high power consumption, large physical footprints, and restricted switching frequencies, especially in the GHz range. This could limit the effective performance and efficiency of MIS. Quantum computing, in turn, today appears as an alternative, at least for fully digital circuits in MIS; QCA provides advantages related to higher intrinsic switching speeds (up to terahertz) compared with transistor-based technologies, along with an improved throughput owing to its inherent compatibility with pipelining. QCA also has minimum power consumption and a smaller area of circuitry, which makes it amply suitable for establishing frameworks in circuit design for AI applications.

The performance requirement in AI is real-time, with minimum energy consumption and minimum cost. The ALU, in this regard, forms the basis for processing and computation units within processor systems. The method presented in this work benefits from the merits of QCA for an ALU design featuring low complexity, high performance, minimum power consumption, maximum speed, and reduced area. This approach has been able to successfully integrate the design of adders and multiplexers with that of basic gates to reduce latency and energy consumption with the aim of improving AI in MIS. The development and simulation of the proposed designs are carefully carried out using QCADesigner 2.0.03 software. A comparison of the different structures proposed shows significant improvements in complexity vs. cell count vs. power consumption compared to earlier designs, hence promising quantum computing for the MIS capability development.

6 QUANTUM IMPLEMENTATION IN MACHINE LEARNING

Machine Learning over a large dataset often requires huge computational power along with a complexity of time. Quantum computing has been used in the domain of ML as well due to its ability to reduce runtime.

6.1 Machine Learning Algorithms for Single Qubit

Considering a superconducting charge qubit or transmon qubit, another ML operation can be shown by the implementation of algorithms based on programmable logic. The model is initialised with a full state vector. The transmon qubits are controlled using microwaves, and their amplitude, phase, and shape determine the resulting quantum state. These operations are called quantum gates.

A Quantum Algorithm follows the regular machine learning processes. Pre-training of data with initialised values from quantum simulations. Quantization-aware training (QAT) is another step that is rather critical for optimizing ML models to perform more efficiently while working with quantum algorithms. The primary motivation behind QAT is to reduce the precision of the weights and activations in neural networks from floating-point to fixed-point representations, thereby decreasing the model size and computational complexity.

6.2 Qiskit: Comprehensive Framework for Quantum Computing

- It is an Open-source software is used as a development tool for quantum information science that eases quantum workflow by addressing problem definition, quantum circuit construction, execution of the hardware or software, and the post-processing of the results obtained.

- Its framework is simple yet comprehensive, supporting both experimental research and production-grade applications.

- A well-appreciated aspect of Qiskit is its granularity, ranging from gate-level operations to real-time classical control flow, which is essential for developing a hybrid quantum-classical model.

- Qiskit is also credited with being able to perform target-specific rewriting and gate-level optimisations for better deployment of quantum machine learning algorithms by a primitive interface with consistent APIs.

- Lastly, it also supports quantum neural networks (QNNs) and dynamic circuits, which lay the base for further complex problems to be undertaken. Coupled with its vast library along with the implementation of high-level algorithms with low-level hardware specifications has guided towards simpler and easier implementation of machine learning applications.

6.3 Conclusion

Quantum Machine Learning algorithms can probe and solve even the toughest and most complex problems comparatively faster than a regular model. Though a new advent, it promises to be exciting and poses to change the landscape of problem solving and data representation.

7 FUTURE OF QUANTUM COMPUTING

7.1 How is quantum computing in the NISQ era?

The advent of Noisy Intermediate-Scale Quantum (NISQ) computing, as introduced by John Preskill in his seminal 2018 paper, marks a pivotal phase in quantum information science. NISQ devices, characterized by 50100 qubits and limited by noise in quantum gates, represent the current frontier of quantum computing technology. Unlike fully fault-tolerant quantum computers, which remain a future goal due to the significant overhead of quantum error correction, NISQ systems operate without robust error mitigation, restricting the size and complexity of executable quantum circuits. Despite these limitations, NISQ technology is poised to explore complex quantum phenomena, particularly in many-body quantum physics, and holds potential for practical applications in optimization, quantum simulation, and machine learning. [8]

Preskill emphasizes that NISQ devices are unlikely to deliver transformative societal impacts immediately but serve as critical stepping stones toward more advanced quantum technologies. The era is defined by the "entanglement frontier," where quantum systems exhibit correlations that classical computers struggle to simulate efficiently. Theoretical arguments, complexity theory, and the absence of efficient classical simulation methods underpin the belief in quantum computational advantages. Key applications include **hybrid quantum-classical algorithms** like the Quantum Approximate Optimization Algorithm (QAOA) and **Variational Quantum Eigensolver (VQE),** which aim to tackle combinatorial and quantum problems, respectively. Additionally, quantum simulation of highly entangled systems offers promise for advancing fields like quantum chemistry and materials science, though significant challenges remain due to noise and limited circuit depth.

The NISQ era also highlights the importance of improving qubit quality, gate fidelity, and noise-resilient algorithm design to extend computational capabilities. Experimental platforms, such as superconducting circuits and trapped ions, are being leveraged for both digital and analog quantum simulations, with the latter currently offering more sophisticated control despite lacking the universality of gate-based systems. Preskill underscores the need for continued research into quantum error correction and hardware advancements to bridge the "quantum chasm" toward fault-tolerant quantum computing. The NISQ era, while constrained, provides a unique opportunity to experiment with quantum hardware, accelerate algorithm development, and lay the groundwork for future quantum breakthroughs.

7.2 Energy-efficient quantum computing

As quantum computers scale, managing thermal dissipation at cryogenic temperatures becomes increasingly difficult due to heat conducted through cabling and dissipated by cryogenic components. The study by Ikonen, Salmilehto, and Möttönen (2017) addresses a fundamental challenge in quantum computing: minimizing energy consumption while maintaining gate fidelity. [4]

As quantum computing progresses toward larger and more complex systems, the challenge of managing thermal dissipation becomes increasingly critical. Quantum processors operate at cryogenic temperatures to minimize decoherence, but as the number of qubits scales into the millions, the heat generated by control cabling alone could reach 10100 mW. This thermal load far exceeds the cooling capacity of current dilution refrigerators, typically limited to approximately 100 ţW at 10 mK. Thus, addressing energy consumption in quantum gate operations is essential for preventing thermal overload and maintaining gate fidelity. (Gate fidelity in quantum computing measures the accuracy of operations, comparing ideal and actual results.)

This work stands apart from earlier studies by introducing a novel approach to energy-efficient gate operations, a topic previously explored in several studies but often with less emphasis on thermal load reduction.

7.3 Previous Studies:

Earlier research on quantum computing focused primarily on improving gate fidelity and fault-tolerant error correction, often at the cost of increasing energy consumption. For instance, studies by Devitt et al. (2013) and Fowler et al. (2012) explored the optimization of surface codes

for error correction but did not address the thermal dissipation due to energy consumption in control pulses. Additionally, Sarovar et al. (2011) examined the energy efficiency of quantum gates but lacked a detailed framework for minimizing the power required for large-scale systems.

7.4 Differentiation of Ikonen et al.'s Work:

Unlike these earlier studies, the work by Ikonen et al. (2017) directly addresses the issue of thermal dissipation in cryogenic environments, particularly the power consumption of control pulses and their impact on cooling systems. Previous models, such as those proposed by Niskanen et al. (2007) and Barends et al. (2014), assumed a direct relationship between gate error and the energy of control pulses, often recommending higher pulse powers to reduce error rates. However, these models did not explore alternatives for minimizing energy without compromising fidelity.

The study begins by identifying the critical issue of heat management in quantum computing. Current quantum error correction protocols, particularly surface codes, require thousands of physical qubits to form a single fault-tolerant logical qubit, leading to substantial heat generation through cabling and attenuation at cryogenic temperatures. The authors assert that scaling up quantum computers necessitates more energy-efficient gate operations to mitigate this thermal load. This context sets the stage for exploring alternative control methods that maintain gate fidelity while reducing power consumption.

The primary contribution of this work is the derivation of a lower bound for the gate error induced by a single bosonic drive pulse of specified energy, using the Jaynes-Cummings model. Unlike previous studies that assumed gate error to be inversely proportional to drive pulse power, Ikonen et al. **propose a qubit-driving scheme that reuses and corrects control pulses using auxiliary qubits.** This approach significantly reduces the average energy consumed per gate operation without increasing the average gate error.

Methodology and Models

- Semiclassical Approach: The authors first review the semiclassical interaction model for single-qubit control, where qubit states are represented on a Bloch sphere and manipulated using classical control pulses in resonance with the qubit transition energy. They derive the average gate error as a function of photon number and introduce a squeezing parameter to minimize this error.
- Quantum Approach: The study then extends to the quantum domain using the Jaynes-Cummings model, which considers the quantum mechanical nature of the control pulse. The authors derive analytical expressions for the lower bound of gate error and identify optimal drive states that minimize this error. Notably, these optimal states are squeezed coherent states that reduce error without increasing energy expenditure

. A significant innovation proposed in the study is the drive-refreshing protocol. In this scheme, an itinerant control pulse interacts with a series of qubits, and auxiliary qubits are employed to correct back-action-induced distortions. This protocol allows for multiple gate operations using a single control pulse while consuming less energy than the conventional approach of using disposable pulses. Simulation results demonstrate that the drive-refreshing protocol can maintain a low average gate error over successive cycles while substantially reducing photon consumption.

The findings challenge the conventional view that increasing pulse power is the only way to reduce gate errors, highlighting that optimized pulse reuse combined with corrective operations can achieve similar or better error rates at a fraction of the energy cost. The study suggests that future quantum processors could leverage such energy-efficient protocols to mitigate the thermal management challenges associated with large-scale quantum systems. Additionally, the authors recommend exploring the extension of these methods to multi-qubit gates and complex quantum operations.

7.5 Conclusion

Ikonen et al. (2017) provide a rigorous analysis of the trade-offs between energy consumption and gate fidelity in quantum computing. By developing a drive-refreshing protocol that corrects control pulses using ancillary qubits, the study presents a promising pathway for reducing power consumption in quantum systems without compromising computational accuracy. The work has implications for the design of scalable quantum processors that operate within the thermal limits of cryogenic environments, potentially advancing the feasibility of large-scale quantum computation.

8 QUANTUM ERROR MITIGATION

Quantum computing today faces a major hurdle: noise. NISQ devices, the current generation of quantum computers, are limited by errors that disrupt computations. While quantum error correction (QEC) offers a long-term solution by encoding logical qubits into multiple physical qubits to detect and fix errors, it demands significant hardware resourcessomething NISQ devices cant yet handle. Enter quantum error mitigation (QEM), a practical alternative for near-term quantum processors. QEM doesnt correct errors during computation but reduces their impact through classical post-processing of noisy results. It uses fewer qubits, avoids the need for ancilla qubits, and is ideal for NISQ systems, though it requires more experimental runs due to increased variance. QEM serves as a stepping stone toward fault-tolerant quantum computing, bridging the gap between todays noisy hardware and tomorrows robust systems.

QEC and QEM differ fundamentally. QEC actively corrects errors in real-time, requiring substantial qubit overhead, making it a future goal. QEM, however, estimates noise-free results after the fact, making it more feasible now, but at the cost of higher sampling needs. Lets explore the key QEM methods and related techniques outlined in the paper.

8.1 Key QEM Methods

The paper highlights several QEM techniques designed to tackle noise in NISQ devices, each with unique strengths and challenges.

8.2 Probabilistic Error Cancellation (PEC)

PEC is a method that counters noise by applying its inverse, effectively cancelling it out. It decomposes the noise into a set of quantum operations weighted by quasi-probabilities (positive or negative coefficients). By running multiple noisy circuits and combining their results via Monte Carlo sampling, PEC reconstructs the ideal, noiseless outcome. Developed by Temme et al. and refined by Endo et al., PEC works for both Markovian and non-Markovian noise and has been tested on superconducting and trapped-ion systems. [3]

However, PECs sampling overhead grows exponentially with noise levels, limiting its efficiency for high-error circuits. Accurate noise modelling is critical. Gate Set Tomography (GST) can characterize noise but is resource-heavy, with sample needs scaling exponentially with system size. For Pauli noise, Cycle Error Reconstruction (CER) is a more efficient alternative, scaling polynomially and demonstrated on small processors. Sparse Pauli-Lindblad models further simplify noise characterization for larger systems.

8.3 Data-Driven Approaches

These methods use empirical data to estimate and correct errors without detailed noise models, making them adaptable to various quantum platforms. [3]

- **Zero-Noise Extrapolation (ZNE):** ZNE runs a quantum circuit at different noise levels (e.g., by increasing circuit depth or gate counts) and extrapolates to a zero-noise result using fitting functions like linear or exponential models. Its versatile, not requiring a detailed noise model, and effective for stochastic errors. However, it assumes predictable noise scaling, which may not hold for complex noise, and requires multiple runs, increasing computational cost.

- **Multi-Dimensional ZNE:** This extends ZNE by considering multiple noise parameters (e.g., gate errors, decoherence times) simultaneously for better extrapolation accuracy. It uses advanced fitting techniques to model noise interactions, improving results for systems with diverse error types. The trade-off is increased modelling complexity and measurement demands.

- **Error Mitigation with Learning Process:** This approach uses machine learning to learn noise patterns from experimental data, predicting error-free outcomes. Neural networks or regression models map noisy outputs to ideal results. It adapts to complex, non-linear noise but requires substantial training data and computational resources, with performance tied to data quality.

8.4 Measurement Error Mitigation (MEM)

MEM targets readout errors, a significant issue in NISQ devices. It assumes classical noise (e.g., bit flips) and uses a calibration matrix to invert errors, built by measuring known states. For quantum-specific errors like state-dependent biases, a more complex noise model is needed. MEM can combine with other QEM methods

(e.g., ZNE) to address both gate and measurement errors. Its straightforward for classical noise but challenging for quantum noise, as calibration matrices grow large with system size.

8.5 Purification-Based QEM Schemes

These methods use extra qubits (ancillae) to purify noisy quantum states, projecting them onto a less noisy subspace. Resource-efficient variants minimize ancilla use, making them NISQ-friendly, while generalized methods handle broader errors, including coherent ones. They can correct stochastic and coherent errors but require deeper circuits, which may introduce new errors. Resource-efficient variants are still being refined to balance accuracy and overhead.

8.6 Quantum Subspace Expansion

Used in variational quantum algorithms like VQE, this method expands the quantum state into a larger subspace to estimate error-free expectation values. Its effective for quantum chemistry applications but limited to specific observables and requires careful subspace selection to manage computational costs.

8.7 Verification Methods

These exploit physical or mathematical constraints, like symmetries, to verify and correct computations. Symmetry verification uses system properties (e.g., particle number conservation) to discard erroneous results, while symmetry expansion incorporates constraints to improve accuracy. Other methods include cross-checking with classical simulations or redundant measurements. Theyre effective when symmetries exist, but less general otherwise.

8.8 Mitigating Coherent Errors

Coherent errors, like systematic over-rotations, preserve state purity but cause inaccuracies. Randomized compiling turns coherent errors into stochastic ones, which are easier to mitigate, while hidden inverses use inverse operations to cancel errors. These methods improve compatibility with other QEM techniques but increase circuit complexity or require precise error knowledge.

8.9 Combination with Quantum Error Suppression

QEM can pair with error suppression techniques for better performance. For example, combining QEM with partial QEC reduces errors in semi-fault-tolerant systems. Dynamic decoupling uses pulse sequences to suppress environmental noise, while single-qubit gate scheduling optimizes timing to minimize error buildup. These approaches enhance QEM but add circuit design complexity and may require advanced hardware control.

8.10 Related Near-Term Quantum Techniques

Beyond QEM, the paper touches on other techniques critical for NISQ devices:

- **Variational Quantum Algorithms (VQAs):** These hybrid quantum-classical algorithms, like VQE, optimize parameters to solve problems like quantum chemistry. Theyre noise-resilient, making them ideal for NISQ systems, but rely on QEM to improve accuracy.

- **Circuit Compilation:** This involves optimizing quantum circuits to reduce gate counts and depth, minimizing noise exposure. Its essential for NISQ efficiency but requires sophisticated software tools.

- **Benchmarking:** Techniques like randomized benchmarking assess quantum processor performance, helping evaluate QEM effectiveness. They provide standardized metrics but can be resource-intensive.

- **Classical Simulation:** Simulating quantum systems on classical computers aids in developing and testing QEM methods. However, its limited to small systems due to exponential complexity.

8.11 Conclusion

QEM and related techniques like variational algorithms, circuit compilation, benchmarking, and classical simulation are vital for advancing NISQ quantum computing. QEM offers a practical way to mitigate noise, with methods like PEC, ZNE, MEM, and others addressing different error types. Each has trade-offs: PEC needs precise noise models, ZNE requires multiple runs, and purification demands extra qubits. Combined with error suppression and other techniques, QEM paves the way for reliable quantum applications, serving as a bridge to future fault-tolerant systems. These advancements are crucial for realizing the potential of quantum computing in the near term.

9 SPACE TECHNOLOGY

Quantum Computing is also heavily linked with a future in space technology, with a few steps already taken. Variational Quantum Circuits (VQCs) are a class of hybrid quantum-classical algorithms that are being utilised to solve supervised learning tasks. These are heavily used space technologies, such as in the identification of Exoplanets

9.1 Architecture of VQC

- A Quantum Map Feature encodes the classical features, such as Kepler Object of Interest (KOI) parameters, into quantum states.

- A parameterized ansatz (initial condition) is optimised through classical methods like COBYLA to minimise the cost function to form a quantum-classical hybrid model.

9.2 Advantages of VQC

- VQCs achieve around 80% accuracy while classifying the exoplanets, emerging superior to other neural networks, given that it requires fewer computational resources due to quantum parallelism driven by entanglement, which means that it implements the functioning of multiple quantum computers on a single algorithm.

- The exhibit enhanced performances while detecting the subtle and noisy transit signals once again, due to quantum parallelism.

- It implements PCA, or Principal Component Analysis, which helps in reducing the number of qubits needed before the quantum encoding. This also aids in reducing quantum noise, leading to higher accuracy.

9.3 Conclusion

This was just a single example that has proven the worth of quantum computing in various fields, even in space technology. With multiple new experiments being conducted, they are also used to study black holes, explore deep space, and even string theory.

10 QUANTUM OPTIMIZATION

The quantum theory provides a proper and systematic mathematical framework to approach a problem through various optimization algorithms that bind the mathematical and physical model together. Over the years, two options have emerged to utilise quantum mechanics for algorithmic optimisation.

10.1 Core Quantum-Inspired Algorithm Design

It focuses on enhancing the performance and exploration of classical algorithmic models using qubits, quantum gates, and quantum entanglement. However, the other model has gained more attention due to its higher performance efficiency.

10.2 Schrödinger-Based Theoretical Modelling

- This approach has become much more favourable among researchers, as it models the optimization process through Schrödingers equation, where the objective function is analogized to the potential energy in a quantum system.

- The probabilistic nature of quantum mechanics aligns with Schrödingers equation, making them naturally compatible with the probabilistic search mechanisms implemented by the advanced optimization algorithms.

- Key operations such as swarm intelligence, weight calculation, and multi-scale evolution can be used quite easily by implementing the solution space as a wave function or a probability distribution.

- By mapping the optimisation problem to a quantum systems potential energy levels, robust mechanisms can venture through a complex, high-dimensional landscape that classical algorithms often struggle with.

Schrödinger Equation for Optimization:
The time-dependent Schrödinger equation models optimization dynamics as:

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + f(x) \right] \psi(x,t)$$

Fig. 3 The objective function is the quantum potential, while the wave function gives the probability distribution of the solutions.

10.3 Conclusion

Quantum algorithms have found deep usage in other quantum applications, most prominently in machine learning. Given their high computational power and low run-time, they are set to be one of the bases for the development of quantum computing in the world of data science.

11 TRANSPORTATION MODELLING

11.1 Quantum Computing in Intelligent transport systems- a survey :

Transportation modelling uses mathematical and computational tools to simulate and optimize the movement of people, vehicles, and goods across networks like roads, railways, and air routes. It accounts for factors such as travel demand, traffic flow, infrastructure capacity, and costs to inform urban planning, reduce congestion, enhance safety, and minimize environmental impacts. This literature review compares how classical and quantum computers approach transportation modelling, highlighting their methods, strengths, limitations, and future potential, with a focus on traffic management, vehicle routing, and autonomous driving applications.

11.2 Classical Computing in Transportation Modelling

Classical computers, relying on binary logic (bits as 0s or 1s) and deterministic algorithms, are the cornerstone of transportation modeling due to their maturity and accessibility.

Methods

- **Data Processing:** Classical systems process large datasets (e.g., traffic counts, GPS, population statistics) using algorithms tailored for tasks like travel demand forecasting or traffic flow simulation.
- **Algorithms:** They employ iterative methods, linear programming, or heuristics (e.g., Dijkstras algorithm for shortest paths) and agent-based simulations (e.g., MATSim). Simulation: Tools like VISSIM, TransCAD, ArcGIS, and Aimsun solve equations representing traffic dynamics, demand patterns, or network constraints.
- **Optimization:** Techniques such as gradient descent or genetic algorithms optimize routes, schedules, and infrastructure investments.

Strengths

- **Mature Ecosystem:** Extensive software libraries and tools support complex modelling tasks. Scalability: High-performance computing (HPC) clusters enable large-scale models.
- **Accuracy:** Reliable for well-defined problems with quality input data.
- **Cost-Effectiveness:** Runs on widely available, affordable hardware.

Limitations

- Computational Complexity: Large-scale or real-time models (e.g., city-wide traffic with millions of agents) are resource-intensive.
- Combinatorial Problems: NP-hard optimization tasks, like traffic signal timing or large-scale vehicle routing, often require approximations due to exponential computation times.
- Uncertainty Handling: Struggles with stochastic systems involving unpredictable variables (e.g., weather, human behavior).

11.3 Quantum Computing in Transportation Modelling

Quantum computing, leveraging principles like superposition, entanglement, and quantum tunnelling, offers the potential to address complex transportation problems more efficiently. Though still experimental, it shows promise in intelligent transportation systems (ITS), particularly in path planning, operation management, and facility layout. [24] [25]

Theoretical Foundations Key quantum algorithms include:

- **Grovers Algorithm:** Provides quadratic speedup for unstructured search, useful for routing and network analysis.
- **Quantum Approximate Optimization Algorithm (QAOA):** Tackles combinatorial optimization problems like traffic signal timing.
- **Shors Algorithm:** Less directly applicable but foundational for network security.

Implementation approaches include universal gate-based quantum computing and quantum annealing (e.g., D-Wave systems). Most applications are hybrid, combining quantum and classical methods due to current hardware limitations.

11.4 Applications

Traffic Management

- **Traffic Flow Optimization:** Researchers have formulated traffic signal control as Quadratic Unconstrained Binary Optimization (QUBO) problems for quantum processors. Hu et al. enhanced the Biham-Middleton-Levine model using Quantum Particle Swarm Optimization (QPSO) for signal timing. Inoue et al. used quantum annealing to transform signal optimization into an Ising Hamiltonian problem, showing scalability for large networks.

- **Traffic Prediction:** Quantum algorithms improve forecasting accuracy. Zhang et al. developed a Quantum Genetic Algorithm-Learning Vector Quantization (QGA-LVQ) neural network for traffic flow prediction. Huang et al. combined Modified Ensemble Empirical Mode Decomposition with Quantum Neural Networks for network traffic prediction. Qu et al.s quantum Spatial-Temporal Graph Convolutional Network outperformed classical methods in congestion prediction.

Vehicle Routing Optimization Quantum approaches excel in combinatorial problems like the Travelling Salesman Problem (TSP) and Vehicle Routing Problems (VRP). Wang et al.s Quantum-Inspired Evolutionary Algorithm for VRP with Time Windows (VRPTW) achieved faster solutions for large instances. Feld et al. and Irie et al. mapped Capacitated VRP (CVRP) to QUBO for D-Waves quantum annealer, reducing travel times by up to 20% and fuel consumption by 15%.

Autonomous Driving Quantum computing enhances:

- **Object Detection:** Baek et al.s quantum convolutional neural networks reduce complexity in object detection.

- **Traffic Image Classification:** Majumder et al.s hybrid classical-quantum model improves resilience to adversarial attacks.

- **Anomaly Detection:** Quantum networks detect rare traffic anomalies in video analysis.

- **Driver Behavior Analysis:** Sheus quantum mechanics-based model captures dynamic driver behaviour.

- **Vehicle Communications Security:** Quantum secret sharing and hybrid neural networks enhance cybersecurity.

Strengths

- **Optimization Speed:** Quantum algorithms offer potential exponential or polynomial speedups for NP-hard problems.

- **Complex Systems:** Efficiently model stochastic systems with many variables.

- **Data Efficiency:** Quantum machine learning may require less data for accurate predictions.

Limitations

- **Technical Constraints:** Current quantum hardware faces decoherence, high error rates, and limited qubit connectivity, necessitating hybrid approaches.

- **Implementation Barriers:** High costs, energy demands for ultra-low temperatures, and specialized expertise limit adoption

- **Experimental Stage:** Most applications are proofs-of-concept, not production-ready.

11.5 Challenges and Future Directions

Quantum computings potential is tempered by significant hurdles:

- **Hardware Scalability:** Overcoming decoherence and increasing qubit counts is critical.

- **Error Mitigation:** Developing robust error-correction techniques is essential.

- **Algorithm Refinement:** Tailoring quantum algorithms for transportation-specific problems requires further research.

- **Accessibility:** Reducing costs and training needs will drive practical adoption.

Future advancements could enable real-time optimization of city-wide traffic flows, significantly reducing congestion, emissions, and fuel use. As quantum hardware matures, hybrid quantum-classical systems will likely bridge the gap, combining quantum optimization with classical data processing.

11.6 Conclusion

Classical computers dominate transportation modelling with their robust, scalable, and cost-effective solutions, but they struggle with computationally intensive and stochastic problems. Quantum computing, while experimental, offers transformative potential for optimizing traffic, routing, and autonomous driving tasks. Its algorithms outperform classical methods in specific cases, but hardware limitations and implementation barriers restrict current applications to proofs-of-concept. Continued research into quantum hardware, algorithms, and hybrid approaches will be crucial to realizing safer, more efficient, and sustainable transportation systems.

12 QUANTUM ARTS AND HUMANITIES

Quantum humanities is an interdisciplinary field exploring how quantum technologies influence and are influenced by human culture, society, and knowledge, guided by the Greek concept of techne (action-oriented process). It bridges quantum mechanics, computer science, the humanities, and social sciences to explore new ways of knowing, creating, and organizing society. Structured around four research areas: Applications, Implications, Reflections, and Development. It investigates how quantum computing reshapes practices, institutional structures, and cultural symbols.

Quantum computing could significantly expedite data analysis in the humanities, particularly in clustering, classification, and pattern recognition, by leveraging quantum machine learning techniques to analyze large datasets more efficiently, uncovering complex patterns that classical algorithms may overlook. In music, quantum phenomena like superposition are used to create compositions where listeners experience different tones, enabling aleatoric music, as seen in Putz and Svozils quantized piano. One fascinating aspect is quantum musical coherent superposition, where classically exclusive musical states can exist simultaneously as a "50:50 quantum g" in the C major scale. In visual arts, projects like Quantum Cinema (20102013) use quantum principles to visualize three-dimensional animated geometry.

In natural language processing, researchers like Coecke et al. develop quantum-native question-answering systems. In law, quantum computing models normative conflicts, such as superposition for legal-illegal scenarios, to optimize decision-making and improve machine learning applications. The MUSE [5] project stands as a pioneering example of quantum humanities in action. This initiative focuses on analyzing film costumes, with a database containing more than 25,000 costume items with detailed attributes such as color, material, design, and condition. By applying quantum machine learning algorithms to this rich dataset, researchers aim to identify abstract costume patterns across different film genres.

The quantum advantage here is significant- quantum computers can process this vast amount of data more efficiently than classical computers, potentially uncovering patterns that would be difficult or impossible to detect using traditional methods

Operating in the noisy intermediate-scale quantum (NISQ) phase, MUSE combines quantum and classical computing, using classical systems to verify quantum results due to error-prone quantum hardware.

Unlike traditional machine learning, which relies on classical computers, quantum approaches like MUSE employ quantum properties like superposition and entanglement to process complex datasets faster or uncover patterns unattainable by classical methods, fostering interdisciplinary collaboration between quantum scientists and humanities researchers. Expanding beyond digital humanities, quantum computing shows potential in cognitive science by simulating complex neural networks and cognitive processes on quantum hardware, achieving faster, more accurate cognitive models that mimic biological neural processes.

By exploiting quantum parallelism, these simulations can represent and process multiple cognitive states simultaneously, aligning with quantum-like cognition research that posits decision-making and perception may exhibit quantum-like properties. Quantum models of cognition leverage superposition and entanglement to represent conflicting mental states or probabilistic decision-making, bridging the gap between biological and artificial intelligence systems and enabling AI to handle nuanced, context-dependent decisions with flexibility unattainable

through classical logic-based systems.

The chapter on **Quantum-Like Cognition and Rationality** presents a particularly intriguing perspective that merits deeper exploration. It discusses how quantum principles such as superposition and entanglement can model cognitive processes like decision-making, perception, and rationality in ways that classical AI systems struggle to replicate. By treating cognitive states as probabilistic rather than deterministic, quantum cognitive models can handle ambiguous, conflicting, or context-dependent information more effectively, potentially leading to AI systems that more closely emulate human thought processes. This quantum approach not only challenges classical logic but also aligns with recent findings in cognitive science, suggesting that the human brain may inherently exhibit quantum-like information processing. Further research in this area could open new pathways for AI development, particularly in systems requiring nuanced, context-sensitive decision-making.

In brain-computer interfacing (BCI), quantum computing enhances systems that establish direct communication pathways between the human brain and external devices, allowing real-time monitoring, control, and feedback. Quantum algorithms process the vast, complex datasets generated by BCIs more efficiently, facilitating sophisticated data encoding and decoding mechanisms to interpret neural signals with greater precision, potentially transforming medical applications like controlling prosthetic limbs through thought or diagnosing neurological conditions. Across these areas, quantum computings potential lies in its capacity to handle complex, high-dimensional data structures, processing multiple data points simultaneously to uncover hidden patterns or correlations more efficiently than classical computers, which process data sequentially.

However, contemporary quantum computers, characterized by limited qubit counts and susceptibility to noise, remain in the NISQ stage, capable of demonstrating proof-of-concept applications but facing significant challenges in scaling to handle real-world data. As quantum technology matures, projects like MUSE [19] could scale to tackle bigger humanities questions, enabling scholars to analyze large, complex datasets with greater depth, revealing new cultural, social, or historical patterns, and preparing the field for a post-NISQ era where quantum computers surpass classical systems in reliability and power.

13 CONCLUSION

This review has covered and emphasized the fast-changing landscape of quantum computing, focusing on its extensive potential across various fields of science and technology, ranging from theory to practicality. In the context of the quantum internet, with huge developments in entanglement-based networks, the future looks very bright for quantum-key-based encrypted fast networks. Quantum computing also presents some major challenges to traditional cryptography, with Shor's and Grover's algorithms endangering asymmetric (RSA, ECC) and symmetric systems respectively. Quantum Key Distribution (QKD), i.e. BB84 and E91, provides secure key exchange but is hampered by hardware vulnerabilities such as Photon Number Splitting and fibre and free-space inefficiencies.

Post-Quantum Cryptography (PQC) offers deployable practical quantum-resistant algorithms (e.g., lattice-based, code-based) but lags behind in terms of large key sizes and computational performance. Although QKD has better security, it is less deployable. Today's developments, which have been implemented by Cambridge, ISRO, and ID Quantique, are aimed at scalability and security, overcoming challenges such as expensive devices and identity verification. Concurrently, quantum computing's disruptive promise in biomedical applications, including drug discovery, genomics, and AI-assisted imaging, is held back by NISQ-era challenges such as noise and scalability.

Methods such as Quantum Error Mitigation (QEM) and hybrid algorithms (QAOA, VQE) boost NISQ performance, whereas innovations such as Ikonen et al.'s drive-refreshing protocol enhance energy efficiency. In transport, quantum algorithms guarantee optimized routing and traffic solutions, and in the humanities, initiatives such as MUSE apply quantum techniques for cultural data analysis and demonstrate how critical breakthroughs in hardware, error correction, and interdisciplinary expertise are needed to unlock quantum technology's potential. Another field that has seen rapid implementation of quantum computing is semiconductors. Quantum simulations have been employed to design and model quantum materials more accurately and precisely than ever, incorporating electronic innovation with the genius of quantum science.

This is leading to developing new device architectures of very small sizes. However, it would not be wrong to sug-

gest that the speed advantage offered by quantum computing has been especially key in algorithmic optimization, leading to more efficient and productive outputs. Its key example is its potential uses in data-driven fields such as machine learning and neural networks, where heavy operations are to be carried out without putting much load on the hardware and keeping time complexity in mind. These complex algorithms are being looked at as a breakthrough in the field of space research as well, emphasising extensive pattern recognition capabilities and heavy algorithmic operations in deep space operations. Thus, quantum computing has found its applications in various fields and promises to elevate them to new heights through rapid developments with the advent of new technology continuously.

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